

Guide to the Perkins Sabre M92

Repair, service and tuning notes for the Perkins Sabre M92 marine diesel, written around one specific engine: serial AR50750U038560F , built in 1999.

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Quick reference: engine and manuals

Everything you need to quote on the phone, and every manual that covers this engine. Detail follows in sections 1 and 2.

The engine

Model	Perkins Sabre M92 (not M92B, a different engine)
Engine number	AR50750U038560F
Second plate number	1396/2400 , shown above the caption TPL No
Type code	AR = Perkins 1004-42 , the base engine
Built	UK, between 1 April and 31 December 1999
Engine	4 cylinder in line, naturally aspirated, direct injection, 4.233 litres, bore 103 mm, stroke 127 mm, 18.5:1
Rating	91 hp at 2400 rpm, firing order 1, 3, 4, 2
Injection pump	Lucas/Delphi rotary, DP200 Series. Confirm from the plate on the pump
Gearbox	PRM 500D or ZF HBW / HSW 450A. Confirm which is fitted

Quote this to a parts desk: AR50750U038560F (TPL 1396/2400).

The manuals

Document	Publication	Covers	Held
Workshop manual	TPD 1350E Iss 4	Engine repair, type AR	Yes
User's handbook	TPD 1397EGN Iss 10	Marine servicing and faults	Yes
Installation manual	TPD 1317E	Install and wiring	Yes
Engine number guide	PP3000/05/15	Serial number decoding	Yes
M92 data sheet	Trans Atlantic Diesels	Ratings and construction	Yes
PRM 500 workshop	Iss 2, June 2000	PRM 500D teardown	Yes
PRM owners handbook	PRM Newage	Gearbox servicing	Yes
ZF HBW repair manual	ZF 310.01.0001	HBW 450 gearbox repair	Yes
DP200 pump chapter	Extract, TPD 1312	Pump removal and timing	Yes
CAV DPA rebuild	Delphi / Lucas CAV	Pump internals. Wrong family	Yes
M92B handbook	N37347	Not your engine	Yes
Illustrated parts book	AR81148 , 116 pp	Part numbers, 1004-42	No

Everything marked Yes is a PDF in [resources/](#) . Exact filenames are listed in [resources/README.md](#) and in section 2.

What is not here, and where to get it

The illustrated parts book, Perkins AR81148. This is the only document class missing. Every copy online sits behind a paywall on a manual reseller. Three ways to get it:

- **Perkins My Engine**, which gives parts information tied to your serial number directly. Start here, it is free
- **A Perkins distributor**, quoting AR50750U038560F in full. Perkins states that end users can order paper service, maintenance and workshop manuals this way
- **Sabre Plant & Marine**, sabrepm.com/downloads, which can supply workshop and parts manuals for older Sabre applications when a local dealer cannot identify the engine

Wiring diagrams are not a separate publication. They are in chapter 7 of the installation manual, from page 37. Yours is Figure 4, M92 insulated return. Figure 3 is the earth return version and Figure 5 is the M92B, so take care which you read.

Injection pump internals are a pump shop job. The DP200 service literature is trade material and is not published openly. The two pump files above cover removal, refitting and timing, which is the work you can do on the engine.

1. Your engine, identified

The photographs show the engine data plate, stamped on a plate at the rear of the right hand side of the cylinder block. That is exactly where the Perkins handbook says to look. Two numbers are visible.





The plate sits low on the block behind the injection pump linkage, partly masked by a bracket and a fuel pipe. Between the two photographs the whole of both numbers can be read.

Serial number: AR50750*U038560F*

Perkins breaks this down as follows (the handbook uses AR30495U123456F as its worked example, so the format matches line for line):

Part	Value	Meaning
Type code	AR	1004-42, the base engine of the M92
Build list	50750	The specific build specification
Country	U	Built in the UK
Serial	038560	Engine serial number
Year	F	Built between 1 April and 31 December 1999

Two independent Perkins documents confirm the type code. The official *Engine Number Guide* lists AR against 1004-42. The M92 user handbook opens its identification section with the flat statement: "M92 engine - identification letters AR". The M115T, the turbocharged sister engine sharing the same handbook, is AK.

On the year letter, note the detail that catches people out: 1999 is split across two codes in the official table. E covers 1 January to 31 March 1999, and F covers 1 April to 31 December 1999. So this engine left the factory in the last three quarters of 1999. Your working assumption of "1999 or 2000" was right, and it lands on 1999.

Second number: 1396/2400

This sits above the caption TPL No on the plate. Perkins does not define "TPL" anywhere in its public literature, and I could not find a source that decodes it, so I am not going to invent a meaning for it. Treat it

as part of the engine's identity and quote it alongside the serial number whenever you order parts or ask a distributor for literature. Perkins is explicit that a finishing or build list shown on the plate must be quoted when ordering spares.

The number to give a parts desk, in full:

AR50750U038560F (TPL 1396/2400)

What this buys you. Knowing the code is AR is the whole game. It means the correct factory repair manual is the *Perkins New 1000 Series, Models AJ to AS and YG to YK Workshop Manual*, publication TPD 1350E. AR sits inside the AJ to AS range. That manual is in [resources/](#), and it carries AR specific data throughout, including a cylinder head tightening procedure written for AR and AS alone. Everything in sections 4 and 9 below is taken from it or from the M92 handbook, not from a generic Perkins source.

One caution carried over from your earlier notes, now confirmed rather than assumed: do not use M92B literature as your repair reference. The M92B is a different engine on the 1100 Series platform. It is worth having for comparison, and a copy is in [resources/](#), but the M92 and the M92B do not share a base engine.

2. Available resources

Downloaded to [resources/](#)

File	What it is	Why it matters
perkins_1000_series_workshop_manual_TPD1350_AJ-AS_YG-YK.pdf	Factory workshop manual, TPD 1350E Issue 4, December 2001, 370 pages	The repair manual for your engine. Covers type AR. Teardown, clearances, torques, timing, overhaul
perkins_sabre_M92_M115T_users_handbook_TPD1397EGN.pdf	Official M92 and M115T user handbook, TPD 1397EGN Issue 10, 96 pages	Marine specific: service schedule, fluids, raw water side, fault charts. English, German, Norwegian
perkins_sabre_marine_installation_manual_TPD1317E.pdf	Perkins Sabre installation manual, TPD 1317E, 94 pages	Cooling, exhaust, shaft, electrical and control installation. Useful when chasing an installation induced fault
perkins_sabre_M92_sales_brochure.pdf	Trans Atlantic Diesels M92 data sheet	One page summary of ratings and construction
perkins_engine_number_guide.pdf	Perkins Engine Number Guide, PP3000/05/15	The official serial decoding tables used in section 1
perkins_sabre_M92B_users_handbook_N37347.pdf	M92B handbook, part N37347, 64 pages	Comparison only. Not your engine
prm_500_workshop_manual.pdf	PRM 500 Workshop Manual, Issue 2, June 2000, 60 pages	Full teardown for the PRM 500D gearbox. Contemporary with your engine

File	What it is	Why it matters
<code>prm_owners_handbook_all_models.pdf</code>	PRM Owners Handbook, 22 pages	Operation and servicing, PRM 80 to 1750
<code>zf_hurth_HBW_360_450_630_repairs_manual.pdf</code>	ZF Marine Repair Manual 310.01.0001, 107 pages	Covers the HBW 450 gearbox. Four languages
<code>perkins_TPD1312_extract_DP200_pump_chapter.pdf</code>	Extract from Perkins TPD 1312, 4 pages	The DP200 pump chapter from the Phaser/1000 manual, including the 100 deg BTDC piston probe timing method
<code>delphi_lucas_CAV_DPA_pump_rebuild_manual.pdf</code>	CAV DPA rebuild manual, 21 pages	Rotary pump internals. Background only, see the caution below

Two cautions on that list. The gearbox manuals cover both options, so check which box you actually have before using either. And the DPA rebuild manual is a different pump family from the DP200 your engine most likely carries: it is useful for understanding how a Lucas rotary pump works, not for working on yours to spec.

Three of the PDFs are also present as extracted plain text (`.txt`) so you can grep them from the command line. For example:

```
grep -n -i "valve tip clearance" resources/*.txt
```

The one gap left

There is no parts catalogue here. The illustrated parts book for the 1004-42 is Perkins publication **AR81148**, 116 pages. Every copy I could find sits behind a paywall on a manual reseller, so it is a purchase rather than a download. Order it against your full engine number, or use Perkins My Engine, which gives parts information tied to the serial directly.

That is the only document class still missing. Engine repair, marine servicing, installation, wiring, both gearbox options and pump removal are all covered by the files above.

Official Perkins sources

- [Perkins marine user guides and manuals](#)
- [Perkins operation and maintenance manuals](#)
- [Find my engine number](#)
- Perkins My Engine app, for parts information tied to your serial
- Wimborne Marine Power Centre (Caterpillar Marine Power UK Ltd), 22 Cobham Road, Ferndown Industrial Estate, Wimborne, Dorset BH21 7PW. This is where Perkins marine engines are engineered, and the M92 handbook is published under their name

Manual libraries

- [ManualsLib: Perkins M92](#)

- [Marine Diesel Basics: Perkins manuals](#)
- [Dieselmann.no manual archive](#) (source of two of the PDFs above)
- [Boatdiesel PDF library](#) (subscription)

Parts suppliers

- [Parts4Engines M92 collection](#). Filters, pumps, impellers, service kits. Lists the Jabsco 29640-1101 bronze sea water pump as original equipment on the M92
- [Trans Atlantic Diesels](#). US based Perkins marine specialist, also the source of the M92 data sheet
- [Lancing Marine](#). Perkins, PRM and Jabsco parts
- [Sabre Plant & Marine downloads](#). Can supply workshop and parts manuals
- [Maritimus](#). European Perkins Sabre spares

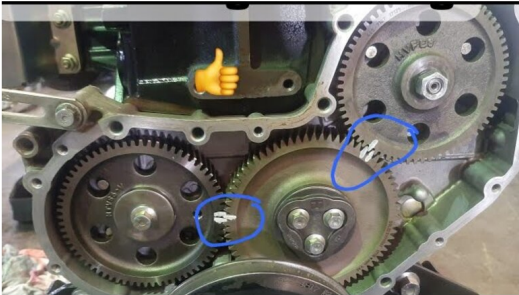
Forums and owner discussion

- [DBA Barge Association: Perkins M92](#). The best single thread on the M92's overcooling problem
- [DBA Barge Association: rattling M92B](#). A cautionary tale, see section 11
- [YBW forum: Perkins Sabre M92 manuals](#)
- [MarineEngine.com Perkins forum](#)
- [Cruisers & Sailing Forums](#), searchable for Perkins injection pump work
- [Attainable Adventure Cruising: M92B report card](#). M92B, but the marine installation lessons carry over

Video

Nobody has published a good M92 specific video series. The base engine is common in agricultural and industrial machines, though, so plenty of 1000 Series and 4.236 footage applies directly to the mechanical work.

Video	What it covers
	Perkins diesel engine timing marks in full HD Shows the injector pump, camshaft, crank and idler gear marks

Video	What it covers
	Installing and timing an injection pump on a Perkins diesel Full removal and refit sequence
	Perkins 4 cylinder diesel injection pump timing Same cylinder count and layout as yours
	Replacing the fuel injection pump, Perkins 1004-4T Closest to your engine family

Also worth reading, though written rather than filmed: [Foley Engines, how to time a Perkins engine.](#)

Watch these for orientation, then do the job from TPD 1350E. The videos will not have your pump variant or your torque figures.

3. What the M92 actually is

A four cylinder, naturally aspirated, direct injection marine diesel of 4.233 litres, rated 91 hp at 2400 rpm. It replaced the 4.236, which had been sold in marine form as the M90, and it was designed to drop into the same installation. Perkins claimed at least 12,000 hours of service life before a basic overhaul.

The two things worth knowing about its construction, because they change how you service it:

Both water pumps are gear driven. The fresh water and sea water pumps run off gears, not a belt. The belt on the front of your engine drives the alternator and nothing else. If the belt snaps you lose charging, but you do not lose cooling. That was a deliberate selling point over the 4.236.

It has an electrical start retard. The injection pump carries a mechanism that holds injection fully advanced for starting and then retards it to the normal running position as the engine warms up. It starts to operate at 55 C coolant temperature. A green lamp visible through a hole in the fuse panel cover tells you it is working. This is not a fault light. Owners who have never read the handbook sometimes chase it as one.

Heat is handled by a fresh water cooled exhaust manifold combined with the inlet manifold and heat exchanger, with a cupro-nickel tubestack. Lubrication runs through an inverted spin-on filter and a disc type oil cooler.

The rating deserves a note. 91 hp at 2400 rpm is a low rated speed for the displacement, and it is why the engine is long lived and quiet. It also means there is no headroom worth chasing. Anyone offering to "tune it up" by winding on fuel is trading engine life for a couple of horsepower you cannot use. Tuning on this engine means restoring it to specification, which is what section 8 is about.

4. Specification and data

Figures below are for the M92 specifically, taken from TPD 1397EGN and from the AR entries in TPD 1350E.

Basic engine

Item	Figure
Cylinders	4, in line, four stroke
Induction	Naturally aspirated
Combustion	Direct injection (Fastram)
Bore	103.00 mm (4.055 in)
Stroke	127.0 mm (5.00 in)
Displacement	4.233 litres (258 in3)
Compression ratio	18.5:1
Firing order	1, 3, 4, 2
Rotation	Clockwise viewed from the front
No. 1 cylinder	At the front of the engine
Maximum power	91 hp at 2400 rpm
Weight, wet, engine only	418 kg (921 lb)
Weight with PRM 500D	499 kg (1100 lb)

Item	Figure
Weight with ZF HSW 450A	455 kg (1003 lb)
Operating angles, continuous	20 deg bow up, 25 deg sideways (35 deg intermittent)

Note the bore and compression ratio. They are the AR and AS signature, and they are how you tell at a glance that a 1000 Series document applies to you. The AJ to AQ engines are 100 mm bore and 17.25:1. If a specification table gives only those figures, it is not describing your engine.

Clearances and pressures

Item	Figure
Valve tip clearance, inlet, hot or cold	0.20 mm (0.008 in)
Valve tip clearance, exhaust, hot or cold	0.45 mm (0.018 in)
Minimum oil pressure, max speed, hot	207 kPa (30 lbf/in ²), 2.1 kgf/cm ²
Compression, expected range	2000 to 3500 kPa (300 to 500 lbf/in ²)
Compression, maximum spread between cylinders	350 kPa (50 lbf/in ²)
Alternator belt deflection at 45 N (10 lbf)	10 mm (3/8 in)

The oil pressure figure is worth committing to memory, because it is lower than most Perkins engines. AJ to AQ and the six cylinder engines want 280 kPa (40 psi). Yours wants 207 kPa (30 psi) minimum at maximum speed, at normal running temperature. A gauge sitting at 32 psi hot on an AR is not a sick engine.

Fuel system

Item	Figure
Injection pump	Lucas rotary with electric and manual stop
Pump type in TPD 1350E	Lucas/Delphi DP200 Series, pin timed, with a locking screw
Governing	Mechanical, all speed
Atomisers	Valve covered orifice (VCO), five hole
Fuel filter	Single element, high mounted, spin-on

The M92 data sheet says "Lucas rotary". TPD 1350E covers three pump families across the 1000 Series: Bosch VE, Lucas/Delphi DP200, and Stanadyne DB2/DB4. Before you touch timing, read the plate on the pump itself and confirm which family you have. The locking screw torque alone differs by a factor of nearly three between the Bosch and DP200 pumps.

Atomiser setting pressures are not one number. Each atomiser carries a two letter code stamped on the body just below the high pressure pipe connection, and the code sets the pressure. The full table is on page 31 of TPD 1350E. Most of the range sits at 290 atm (4263 lbf/in²), but codes run from 250 to 300 atm, so read your code rather than assuming.

Electrical

Item	Figure
12V system battery	One 12V, 520 A to BS3911
24V system battery	Two 12V, 440 A to BS3911
Start retard operates at	55 C (131 F) coolant

Gearbox

Gearbox	Oil capacity	Oil specification
Newage PRM 500D	2.5 litres (4.40 pints)	Engine oil, API CD or ACEA E2
ZF HSW 450A	2.0 litres (3.52 pints)	ATF

Both capacities exclude the oil cooler and its pipes, and both vary with installation angle. Fill to the dipstick, not to the book figure.

For anything past an oil change, use the gearbox manual rather than this guide.

[prm_500_workshop_manual.pdf](#) covers the PRM 500D down to component level, and

[zf_hurth_HBW_360_450_630_repair_manual.pdf](#) covers the HBW 450. Establish which gearbox you have first: the two are unrelated designs and use different oil, engine oil for the PRM and ATF for the ZF.

5. Fluids and capacities

Item	Capacity
Lubricating oil, including filter	8.5 litres (16.5 pints)
Lubricating oil, sump only	7 litres (15.8 pints)
Coolant, engine only	8.75 litres (15.4 pints)

Sump capacity changes with the angle the engine sits at. Never fill past the "Full" mark on the dipstick regardless of what the table says.

Engine oil. Naturally aspirated engines should use API CD or ACEA E1 as a minimum. API CF4 or ACEA E2 can be used, but Perkins does not recommend it during the first 20 to 40 hours, nor for light load work. Match viscosity to your ambient temperature range using the chart on page 61 of the handbook.

Fuel. Cetane 45 minimum. Viscosity 2.5 to 4.5 centistokes at 40 C. Density 0.835 to 0.855 kg/litre. Sulphur 0.2 % by mass maximum. Distillation 85 % at 350 C. Fuel sulphur content changes the oil change interval, so if you are buying fuel of unknown quality, shorten the interval rather than argue about it.

Coolant. Renew antifreeze every two years. If you use a coolant inhibitor instead of antifreeze, renew it every six months. Perkins is firm that antifreeze concentration must be maintained rather than topped up with plain water.

6. Maintenance schedule

From TPD 1397EGN. Apply whichever interval comes first, hours or months.

- **A** First service at 25/50 hours
- **B** Every day or every 8 hours
- **C** Every 250 hours or 12 months
- **D** Every 500 hours or 12 months
- **E** Every 1000 hours
- **F** Every 2000 hours

Operation	A	B	C	D	E	F
Check coolant level in header tank		x				
Check engine for oil and coolant leaks		x				
Check specific gravity of coolant				x		
Check tension and condition of drive belt	x		x			
Check impeller of raw water pump				x		
Check sea water strainer		x				
Clean sediment chamber and strainer of fuel lift pump				x		
Drain water from fuel pre-filter	x	x				
Renew fuel filter element				x		
Atomiser maintenance						x

Operation	A	B	C	D	E	F
Check and adjust idle speed	x					
Check oil level in sump		x				
Check oil pressure at the gauge		x				
Renew engine lubricating oil			x			
Renew lubricating oil filter canister			x			
Check oil level in reverse gearbox		x				
Renew oil in reverse gearbox	x		x			
Renew engine breather						x
Renew air filter element				x		
Check all hoses and connections				x		
Check valve tip clearances, adjust if needed	x				x	
Check audible warning system				x		
Check alternator, starter motor etc.					x	
Check engine mounts				x		

Operation	A	B	C	D	E	F
Inspect electrical system for damage					x	

Jobs marked in the handbook as requiring trained personnel: coolant specific gravity, atomiser work, idle speed, breather renewal, valve clearances, alternator and starter checks, electrical inspection.

The engine breather assembly is renewed complete at major service or 8000 hours.

7. Routine service jobs

Raw water pump impeller

Due every 500 hours or 12 months, and the single most common reason an M92 stops in the middle of a passage.

1. Close the seacock. Confirm it, do not assume it.
2. Release the six setscrews holding the end plate and remove the plate. Some raw water will run out.
3. Remove the rubber end cap and pull the impeller off the shaft.
4. Clean the contact faces of the pump body and end plate.
5. Inspect the impeller for wear and damage. Renew it if in doubt. **If blades have broken off, you must find them.** Remove the outlet hose from the raw water pump, remove the end cap of the gearbox oil cooler, clear the debris, and check that the cooler tube ends are open. Refit the hose and end cap, tighten the clips, and top up the coolant circuit.
6. Apply Spheerol SX2 grease or liquid soap to the blades. Fit the impeller into the housing **with the blades bent clockwise**. Refit the rubber end cap.
7. Apply POWERPART jointing compound, part 1861117, to a new joint. Fit it with the wide area of the joint over the eccentric plate in the body. Fit the end plate and tighten.
8. Open the seacock.

Blade direction matters and the two handbooks in [resources/](#) describe it in different words. The figure above is what your manual, TPD 1397EGN, says for the M92. Carry a spare impeller and a spare end plate joint aboard, and check the cam for wear while you are in there.

The pump itself is a Jabsco 29640-1101 bronze unit as original equipment.

Fuel filter

Due every 500 hours or 12 months.

Lubricate the two top seals of the new canister with clean fuel. Fit to the filter head and tighten **by hand, half to three quarters of a turn**. Then bleed the system.

The handbook carries a specific warning here: the canister seals must be fitted correctly or the injection pump can be damaged. Some canisters ship with a seal held in place by a plastic clip. Check it before you spin the filter on.

Fuel pre-filter

Normally fitted between tank and engine. Check the bowl for water regularly and drain as needed. This is a daily job in the schedule and takes ten seconds.

Drive belt

Press the belt at the centre of its longest free run with moderate thumb pressure, 45 N (10 lbf). Correct deflection is 10 mm (3/8 in).

To adjust: loosen the alternator pivot fastener and the two setscrews on the adjustment link, move the alternator, retighten, then check again. A new belt must be rechecked after the first 25 hours.

Use a Perkins POWERPART belt. The alternator drive uses a belt of a specific section, and a generic substitute fails early.

Finding a bad atomiser

Perkins does not schedule atomiser maintenance except at 2000 hours, and says nozzles should be renewed rather than cleaned, and only when a fault appears.

To find which one is at fault, run the engine at fast idle. Loosen and retighten the high pressure pipe union nut at each atomiser in turn. When you loosen the nut on a defective atomiser, engine speed barely changes.

High pressure fuel will penetrate skin. If it does, get medical help immediately, do not wait to see how it develops.

8. Tuning and adjustment

Tuning an M92 means getting it back to the numbers in section 4. There is nothing else to gain.

Valve tip clearances

Inlet 0.20 mm (0.008 in), exhaust 0.45 mm (0.018 in), **hot or cold**. That is a real convenience of this engine: you do not need to catch it at a particular temperature.

Clearance is measured between the top of the valve stem and the rocker lever. The valve sequence from number 1 cylinder at the front is:

Cylinder	1	1	2	2	3	3	4	4
Valve no.	1	2	3	4	5	6	7	8
Type	I	E	I	E	I	E	I	E

The four cylinder procedure, TPD 1350E Operation 3-6, uses the rocking pair method:

1. Turn the crankshaft in the normal direction until the inlet valve of number 4 (valve 7) has just opened and the exhaust valve of number 4 (valve 8) has not fully closed. Check and adjust valves 1 and 2 (cylinder 1).
2. Set number 2 rocking (valves 3 and 4). Check and adjust valves 5 and 6 (cylinder 3).
3. Set number 1 rocking (valves 1 and 2). Check and adjust valves 7 and 8 (cylinder 4).
4. Set number 3 rocking (valves 5 and 6). Check and adjust valves 3 and 4 (cylinder 2).

Adjusting screws on early engines take a screwdriver, later ones a female Torx. The two types interchange, so do not be alarmed if a replacement part looks wrong.

Rocker shaft bracket fasteners: 40 Nm (30 lbf ft) for aluminium brackets, 75 Nm (55 lbf ft) for cast iron or sintered steel. Getting this wrong distorts the shaft. Identify your bracket material before you reassemble.

Rocker cover cap nuts: 20 Nm (15 lbf ft) for a composite plastic cover, 30 Nm (22 lbf ft) for aluminium.

Idle speed

The schedule asks for idle to be checked and adjusted at the first service, by someone trained. Beyond that it is left alone. Adjustment is at the injection pump, and on a marine installation you also need to confirm the throttle cable is not preloading the lever.

Both maximum no-load speed and idle live on the pump. The maximum no-load speed for your engine is stamped on the emissions data plate on the left side of the cylinder block. That is the number to work to, not the rated 2400 rpm.

Injection pump timing

Do not start here. Timing rarely drifts on its own. It changes because someone removed the pump, or fitted a different gasket, or disturbed the hub.

If you do need to go in, the essentials from TPD 1350E:

- The engine must be at TDC on number 1, compression stroke, before the pump goes on.
- **Do not slacken the nut on the pump hub.** The hub is set to the shaft at the factory to put the pump in the right position for timing. Move it, and it has to be reset with equipment only Perkins distributors have.
- Do not rotate the crankshaft with the pump off the engine. The loose pump gear can damage the timing case. If you must turn the engine, refit the pump temporarily, and when you do, release the locking screw and fit the spacer under it so the pump shaft is free.
- The DP200 pump is pin timed with a locking screw. Locking screw torque is 10 Nm (7 lbf ft) for a DP200, and 27 Nm (20 lbf ft) for a Bosch VE. Use the right one for your pump.
- Fuel pump gear fasteners: 28 Nm (20 lbf ft), setscrew or Torx. On engines with a belt driven coolant pump a special Torx socket is needed to get at them.
- High pressure pipe union nuts: 22 Nm (16 lbf ft).

- Inspect and if needed renew the O ring in the pump flange, and lubricate it with clean engine oil before fitting.

A well known trap on rotary pumps: the O rings inside the pump on each injector body harden with age and change behaviour. That is a pump shop job, not a dockside one.

Compression testing

Set the valve clearances first. Then remove the atomisers, fit a gauge into an atomiser hole, disconnect the stop solenoid or put the stop control in the no-fuel position so the engine cannot start, and crank.

Expect somewhere in 2000 to 3500 kPa (300 to 500 lbf/in²). The absolute number means little, because battery condition, ambient temperature and gauge type all move it. What matters is the spread across the four cylinders. More than 350 kPa (50 lbf/in²) between cylinders points at the low one.

Perkins is blunt about this: a compression test on its own does not tell you the condition of an engine. Use it with other symptoms.

9. Torque figures

Cylinder head, engine types AR and AS

Your engine uses Operation 3-14, which is written specifically for AR and AS. AJ to AQ engines use a different procedure. Do not follow a 1000 Series head tightening sequence that does not name your type code.

You will need an angle gauge, Perkins part 21825607, or you can mark the head and fasteners and work to the marks.

1. Clean the head and block faces. Nothing in the bores.
2. Both location pins must be pressed into the block **before** the head goes on, or the head joint is damaged.
3. Fit the head joint dry. **No jointing compound.** It is stamped "FRONT TOP" for orientation.
4. Fit two M10 guide studs in positions 16 and 21, place the head, and make sure both location pins engage fully.
5. Lightly oil the setscrew threads and the underside of the heads. The four 1/2 UNF setscrews go in positions 2, 8, 13 and 18.
6. Tighten all setscrews gradually and evenly to **45 Nm (33 lbf ft)** in sequence. Repeat the pass to confirm.
7. Tighten the four 1/2 UNF setscrews to **110 Nm (80 lbf ft)** in the order 2, 8, 13, 18. Repeat to confirm.
8. Angle tighten in sequence, by setscrew length:

Setscrew	Further rotation
Short M10 (S)	120 deg (2 flats)

Setscrew	Further rotation
Medium M10 (M)	120 deg (2 flats)
Long M10 (L)	150 deg (2.5 flats)
1/2 UNF, positions 2, 8, 13, 18	180 deg (3 flats)

Reset the valve clearances afterwards.

Specific torques

Lightly oil fasteners with clean engine oil before fitting.

Component	Thread	Nm	lbf ft
Rocker shaft brackets, aluminium	M12	40	30
Rocker shaft brackets, cast iron / sintered	M12	75	55
Rocker cover cap nuts, plastic cover	M12	20	15
Rocker cover cap nuts, aluminium cover	M12	30	22
Inlet manifold to head	M10	44	33
Inlet manifold to head	M8	22	16
Exhaust manifold to head	M10	33	24
Engine lift bracket	M10	44	33
Connecting rod nuts	1/2 UNF	125	92
Connecting rod setscrews	1/2 UNF	152	114
Piston cooling jet banjo bolts	3/8 UNF	27	21
Main bearing setscrews	5/8 UNF	265	196
Crankshaft pulley setscrews	7/16 UNF	115	85
Rear oil seal housing to block	M8	22	16
Timing case to block	M8	22	16
Timing case to block	M10	44	33
Idler gear hub	M10	44	33
Camshaft gear setscrew	M12	95	74
Timing case cover to case	M8	22	16
High pressure fuel pipe nuts	M12	22	16

Component	Thread	Nm	lbf ft
Leak-off banjo bolt	M8	9	7
Atomiser body gland nut	-	40	30
Fuel injection pump gear setscrews / Torx	M10	28	20
Fuel lift pump setscrews	M8	22	16
Fuel injection pump flange nuts	M8	22	16
Locking screw, DP200 pump	10 A/F	10	7
Locking screw, Bosch VE pump	M10	27	20
Sump drain plug	3/4 UNF	34	25
Oil pump to front bearing cap	M8	22	16
Oil pump cover	M8	28	21
Sump fasteners	M8	22	16

Standard torques, setscrews and nuts

Thread	Nm	lbf ft
M6 x 1.00	9	7
M8 x 1.25	22	16
M10 x 1.50	44	33
M12 x 1.75	78	58
M14 x 2.00	124	91
M16 x 2.00	190	140

Standard torques, pipe unions, plugs and adaptors

Thread	Nm	lbf ft
1/8 PTF	9	7
1/4 PTF	17	13
3/8 PTF	30	23
3/4 PTF	45	35

10. Fault diagnosis

The handbook splits causes into what an owner can check and what belongs in a workshop. Numbers below match TPD 1397EGN pages 64 and 65.

Symptom to cause

Problem	Owner checks	Workshop checks
Starter turns engine too slowly	1, 2, 3, 4	
Engine does not start	5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 17	34, 35, 36, 37, 38, 42, 43, 44
Engine hard to start	5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19	34, 36, 37, 38, 40, 42, 43
Not enough power	8, 9, 10, 11, 12, 13, 16, 18, 19, 20, 21	34, 36, 37, 38, 39, 40, 41, 43, 63
Misfire	8, 9, 10, 12, 13, 15, 20, 22	34, 36, 37, 38, 39, 40, 41, 43
High fuel consumption	11, 13, 15, 17, 18, 19, 22, 23	34, 36, 37, 38, 39, 40, 42, 43, 44, 63
Black exhaust smoke	11, 13, 15, 17, 19, 21, 22	34, 36, 37, 38, 39, 40, 42, 43, 44, 63
Blue or white exhaust smoke	4, 15, 21, 23	36, 37, 38, 39, 42, 44, 45, 52, 58, 61, 62
Oil pressure too low	4, 24, 25, 26	46, 47, 48, 50, 51, 59
Engine knocks	9, 13, 15, 17, 20, 22, 23	36, 37, 40, 42, 44, 46, 52, 53, 60
Engine runs erratically	8, 9, 10, 11, 12, 13, 15, 16, 18, 20, 22, 23	36, 38, 40, 41, 44, 52, 60
Vibration	13, 18, 20, 27, 28	36, 38, 39, 40, 41, 44, 52, 54
Oil pressure too high	4, 25	49
Oil temperature too high	11, 13, 15, 19, 27, 29, 30, 32, 65, 66, 67, 68	34, 36, 37, 39, 52, 55, 56, 57, 69
Crankcase pressure	31, 33	39, 42, 44, 45, 52, 61
Bad compression	11, 22	37, 39, 40, 42, 43, 44, 45, 53, 60
Engine starts and stops	10, 11, 12	

Gearbox:

Problem	Owner checks	Workshop checks
Delayed gear engagement	70, 71	
No transmission	72	75
Boat will not reach maximum speed	73, 74	75, 76, 77

Cause list

1. Battery capacity low

2. Bad electrical connections
3. Fault in starter motor
4. Wrong grade of lubricating oil
5. Starter motor turns engine too slowly
6. Fuel tank empty
7. Fault in stop solenoid, contacts or cables
8. Restriction in a fuel pipe
9. Fault in fuel lift pump
10. Dirty fuel filter element
11. Restriction in air induction system
12. Air in fuel system
13. Faulty atomisers, or atomisers of an incorrect type
14. Cold start system used incorrectly
15. Fault in cold start system
16. Restriction in fuel tank vent
17. Wrong type or grade of fuel
18. Restricted movement of engine speed control
19. Restriction in exhaust pipe
20. Engine temperature too high
21. Engine temperature too low
22. Incorrect valve tip clearances
23. Too much oil, or wrong oil, in wet type air cleaner if fitted
24. Not enough oil in sump
25. Defective gauge
26. Dirty lubricating oil filter element
27. Fan damaged
28. Fault in engine mounting or flywheel housing
29. Too much oil in sump
30. Restriction in air or water passages of radiator
31. Restriction in breather pipe
32. Insufficient coolant in system
33. Vacuum pipe leaks, or fault in exhauster
34. Fault in fuel injection pump
35. Broken drive on fuel injection pump
36. Timing of fuel injection pump is incorrect
37. Valve timing is incorrect
38. Bad compression

39. Cylinder head gasket leaks
40. Valves are not free
41. Wrong high pressure pipes fitted
42. Worn cylinder bores
43. Leakage between valves and seats
44. Piston rings not free, worn or broken
45. Valve stems or guides worn
46. Crankshaft bearings worn or damaged
47. Lubricating oil pump worn
48. Relief valve does not close
49. Relief valve does not open
50. Relief valve spring broken
51. Fault in suction pipe of lubricating oil pump
52. Piston damaged
53. Piston height incorrect
54. Flywheel housing or flywheel not aligned correctly
55. Fault in thermostat, or thermostat of incorrect type
56. Restriction in coolant passages
57. Fault in water pump
58. Valve stem seal damaged
59. Restriction in sump strainer
60. Valve spring broken
61. Breather assembly worn or broken
62. Vent hole for breather valve restricted
63. Leakage in the induction system
64. Spare
65. Drive belt for water pump loose
66. Restriction in the seacock or the raw water strainer
67. Insufficient coolant in circuit
68. Restriction in the heat exchanger or the oil cooler
69. Fault in raw water pump
70. Movement of gearbox control lever not equal both ways
71. Insufficient movement of gearbox control cable
72. Gearbox control cable not free, radii too small, or broken
73. Wrong type of oil in reverse gearbox
74. Gearbox oil cooler needed for the conditions of operation
75. Worn or broken drive components

76. Propeller wrong size or not matched

77. Propeller damaged

11. Known weak points on this engine

Four things come up again and again in owner discussion. None of them are in the manuals, which is exactly why they are worth writing down.

Overcooling, not overheating

The M92 has a reputation for running cold rather than hot, particularly at low revs. Owners on the DBA barge forum report engines that reach 80 C and then sag back, or never get there at all below about 1400 rpm. Above that the problem disappears.

Suspects, in the order worth checking:

- Thermostat stuck open, or a thermostat of the wrong rating. This is cause 55 in the fault list and it is the usual answer. One owner ended up replacing the whole header tank and exhaust manifold casting. Others have fitted an aftermarket 87 degree stat successfully.
- The plastic retaining ring not seating properly, letting water bypass the thermostat.
- Heat being taken off by a calorifier or cabin heating circuit faster than a lightly loaded engine can make it.
- On keel cooled installations, more cooling surface than the engine needs.
- No engine insulation on a light load installation.

A diesel run cold for long periods bores and glazes. If yours will not hold temperature, treat it as a real fault rather than a quirk.

Corroded electrical connections that look mechanical

One M92B owner spent three months and six engineers, including a Perkins specialist, on a rattling, trembling, stalling engine with poor throttle response. Everyone assumed injection or a bearing. The cause was a small corroded electrical plug. An electrician found it in short order.

The lesson generalises to every marine diesel, and it belongs at the top of your list, not the bottom. Before you pull anything apart, unplug, inspect and reseal every connector you can reach. Look especially at the stop solenoid circuit (cause 7) and the start retard wiring. Salt air gets into plugs and produces symptoms that read as mechanical.

The circuit diagram you need is in the installation manual, not the handbook. TPD 1317E chapter 7 starts at page 37 and lists a diagram per model. Yours is **Figure 4, M92 insulated return**. Do not use Figure 3, which is the earth return version, or Figure 5, which is the M92B. The same chapter covers the engine wiring loom, the multi-way connector, the fuse and circuit breaker ratings, the interconnecting cables and the Y harness for multiple panels, and it carries separate circuit diagrams for the instrument and control panels.

Starter solenoid contacts

Reported on marineengine.com: a copper disc inside the solenoid making poor contact, giving a starter that turns the engine at about half speed. That reads as a flat battery or a bad earth, and it is neither. Worth knowing before you buy a starter motor.

Impeller debris in the gearbox oil cooler

Covered in section 7, but it belongs here too. When an impeller sheds blades on this engine, the pieces travel to the reverse gearbox oil cooler. Fitting a new impeller without going after the fragments leaves you with a cooling restriction and a fault that appears weeks later, by which time nobody connects the two.

12. Ordering parts

Quote the full engine number every time. Perkins says so in the handbook, and distributors will ask.

AR50750U038560F (TPL 1396/2400)

Take a photograph of the plate and keep it on your phone. The plate in the photographs above is already partly obscured by a bracket and a fuel pipe, and it will not get easier to read.

Also useful to have ready when you call:

- Gearbox model and ratio (PRM 500D or ZF HSW/HBW 450A)
- Whether the panel is 12V or 24V
- Photographs of the injection pump side and of the heat exchanger and exhaust side

For the injection pump specifically, read and photograph the plate on the pump itself. Pump part numbers do not follow from the engine serial, and getting the wrong family wastes a trip.

Known original equipment:

Part	Detail
Sea water pump	Jabsco 29640-1101, bronze
Cooling system cleaner	POWERPART Easy Flush, 21825001
Raw water pump joint compound	POWERPART jointing compound, 1861117
Gasket and flange sealant	POWERPART, 21820518
Gasket remover	POWERPART aerosol, 21820116
Head setscrew angle gauge	Perkins 21825607
Bosch pump flange nut spanner	Perkins 21825964
Injection pump timing pin	Perkins 27610032

Filter service kits covering air, fuel and oil for the M92 are sold as a set, which is the easy way to buy them.

13. Laying up and winterising

Chapter 7 of TPD 1397EGN covers preservation properly. The part people skip is the raw water side.

To put antifreeze through the raw water system:

1. Run the engine, drawing an antifreeze and water mix in through the raw water strainer from a container, swapping containers as they empty.
2. Once the mixture has circulated fully, stop the engine and refit the top of the raw water strainer.
3. Reconnect the hose to the raw water connection on the exhaust elbow.
4. **Put a label on the engine** telling whoever starts it next that there is antifreeze in the raw water system and it must be drained before the seacock is opened and the engine run.

That label is not decoration. The person who starts the engine in spring may not be you.

Renew antifreeze in the fresh water circuit every two years. If you use an inhibitor instead, every six months.

14. Safety

Short list, and all of it is in the handbooks.

- High pressure fuel penetrates skin. If it gets you, get medical help immediately. Do not wait.
 - Stay clear of moving parts with the engine running. Several moving parts on a running diesel are effectively invisible.
 - Before any compression test or cranking work, disconnect the stop solenoid or set the stop control to no-fuel so the engine cannot start.
 - Close the seacock before opening the raw water pump, and check that you actually closed it.
 - Some seals on this engine are Viton. Burnt Viton produces hydrofluoric acid residue. TPD 1350E has a section on handling it. Read it before you take a torch to anything.
 - Never fill past the "Full" mark on the dipstick.
-

Sources

- Perkins New 1000 Series Workshop Manual, Models AJ to AS and YG to YK, TPD 1350E Issue 4, December 2001
- Perkins M92 and M115T Marine Diesel Engines User's Handbook, TPD 1397EGN Issue 10, March 2013, Wimborne Marine Power Centre
- Perkins Sabre Marine Engines Installation Manual, TPD 1317E
- Perkins Engine Number Guide, publication PP3000/05/15
- PRM 500 Workshop Manual, Issue 2, June 2000, PRM Newage

- ZF Marine Repair Manual, HBW 360 / 360A / 450 / 630, document 310.01.0001
- Extract from Perkins Workshop Manual TPD 1312, Lucas/Delphi DP 200 Series chapter
- [Trans Atlantic Diesels: Perkins Sabre M92 data sheet](#)
- [Parts4Engines: Perkins M92 parts](#)
- [DBA Barge Association: Perkins M92 thread](#)
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- Engine data plate photographs, [images/engine-plate-closeup.jpg](#) ,
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